

Unit 6, Lesson 6: Graphing Two-Variable Linear Inequalities for a Real-World Context



Benchmarks

MA.912.AR.2.8 Given a mathematical or real-world context, graph the solution set to a two-variable linear inequality.

MA.912.AR.9.6 Given a real-world context, represent constraints as systems of linear equations or inequalities. Interpret solutions to problems as viable or non-viable options.



Learning Targets

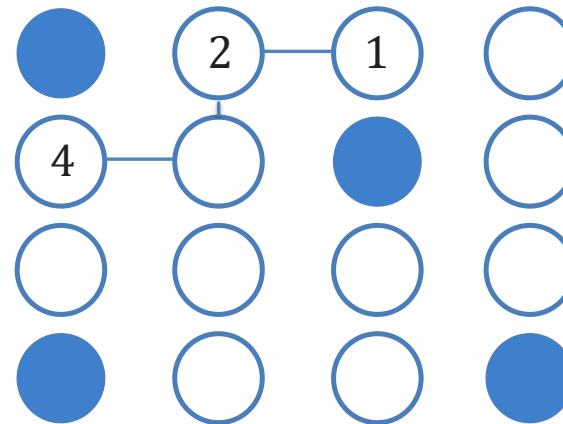
- ☐ I can graph the solution set of a real-world linear inequality, given in standard form, slope-intercept form, or point-slope form on a coordinate plane.
- ☐ I can relate the key features of the inequality to the context.
- ☐ I can interpret solutions to problems as viable or non-viable options.



6.6.1 Warm-Up: Hebi Puzzle

Hebi is a Japanese number puzzle. The word Hebi means snake. The goal is to place number snakes on the grid so that each snake connects the digits with consecutive numbers joining vertically or horizontally to adjacent circles. Numbers never connect diagonally. Snakes cannot cross blue circles.

1. Use the numbers 1 – 4 to fill in the empty circles so that a string of numbers can be connected in numerical order.





6.6.2 Collaboration: U-Pick Berries

5K Farms is an organic berry farm in Plant City, Florida. In the spring, 5K Farms has U-Pick strawberries for \$2 per pound and blackberries for \$5 per pound. Aaron is going berry picking and has \$20.

1. Determine whether each of the following scenarios is possible given the information above.

Scenario	Cost	Is this possible?
Aaron picks 3.2 pounds of blackberries and 1.9 pounds of strawberries.		☐ Yes ☐ No
Aaron picks 2.7 pounds of blackberries and 3.8 pounds of strawberries.		☐ Yes ☐ No
Aaron picks just 4.5 pounds of strawberries and no blackberries.		☐ Yes ☐ No

2. Compare your responses with your partner and resolve any differences.

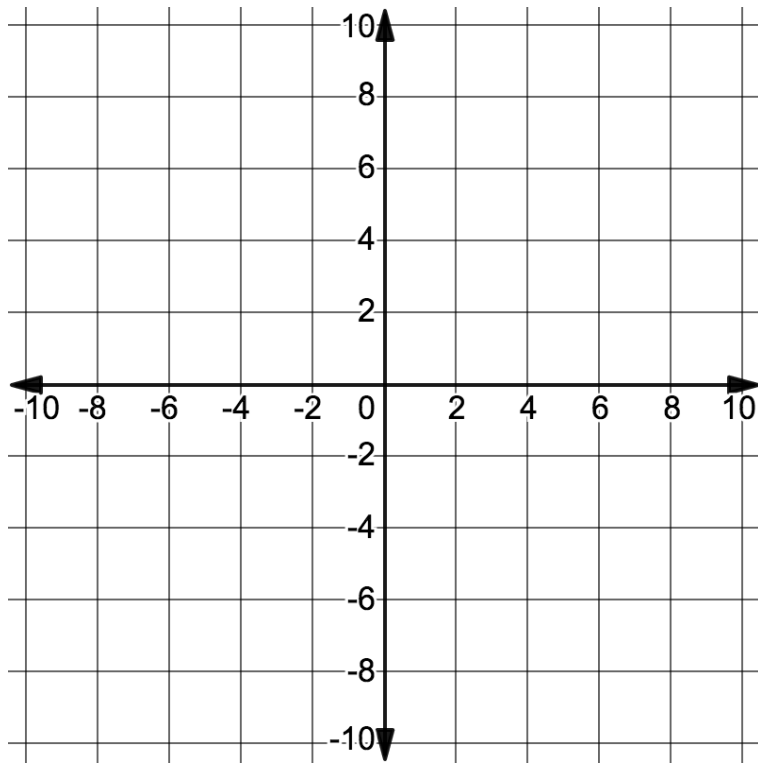


6.6.3 Guided Instruction: Connecting Real-World Contexts and Two-Variable Linear Inequalities

Aaron's scenario can be represented by a linear inequality in two variables.

1. To determine the combinations of each type of berry he could buy, Aaron wrote the inequality $2x + 5y \leq 20$.
 - a. Define the variables Aaron used based on the context.
Let x represent _____
Let y represent _____
 - b. Explain why the inequality sign $\leq$ makes sense in this context.

- c. Graph the inequality $2x + 5y \leq 20$.



- d. The ordered pair $(-1, 2)$ is a solution to the inequality $2x + 5y \leq 20$. Explain why this is not a viable solution in the context of buying berries.
- e. With your partner, discuss which solutions to the mathematical inequality are not viable solutions in this context.

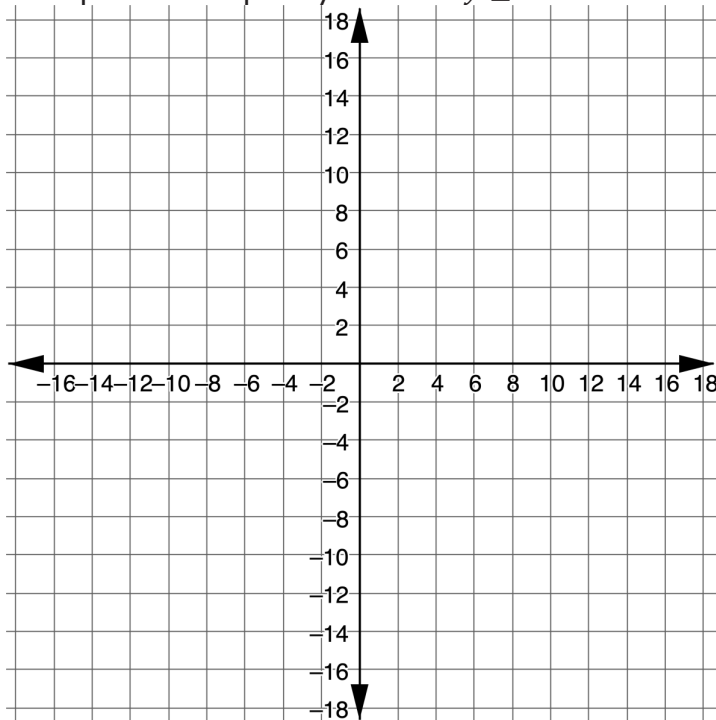
- f. Select the additional constraints necessary on x and y in order for a solution to be viable in the berry-picking context.
- ☐ $x > 0$
 - ☐ $x \geq 0$
 - ☐ $y > 0$
 - ☐ $y \geq 0$
- g. The ordered pair $(4.3, 2.28)$ is a point on the boundary line. Substitute the ordered pair into the inequality $2x + 5y \leq 20$.
- h. Explain what ordered pairs on the boundary line represent in the situation.



6.6.4 Guided Practice: Two-Variable Linear Inequalities in Real-World Contexts

1. Jolene invested some of her savings in the stock market. She bought 20 shares of a tech company and 35 shares of an automotive company. She hopes to make a combined profit of at least \$280 when she sells her shares. She wrote the inequality $20x + 35y \geq 280$ to represent the situation, where x is the change in value of one share of the tech company and y is the change of value of one share of the automotive company.
 - a. Explain what it would mean if x or y were a negative number.

- b. Graph the inequality $20x + 35y \geq 280$.



- c. Explain what the point $(-2.25, 15.75)$ represents in Jolene's situation.
 - d. Explain why it makes sense in Jolene's context that there are no solutions for which both x and y are both negative.

Stock prices in the United States are rounded to the hundredths place.

- e. Complete the statement.

In order for solutions to the mathematical inequality to be viable solutions in the real-world context, the x and y values must be

- ☐ greater than 0.
- ☐ non-negative.
- ☐ integers.
- ☐ rounded to ____ decimal places.

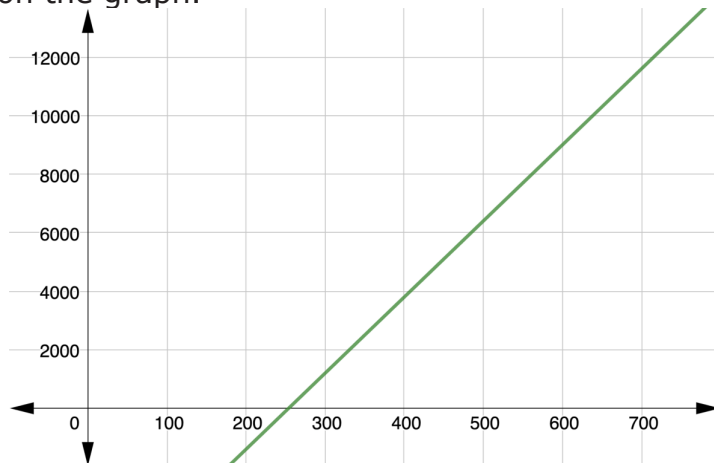
- f. Explain what a solution on the boundary line of the inequality would mean in the real-world context.



6.6.5 Try It: Representing Real-World Contexts With Two-Variable Inequalities

1. In the 2018-2019 harvest year, Florida produced approximately 43.7% of the total citrus produced in the United States. Given the acreage of land and data collected by the Florida Department of Agriculture and Consumer Services, citrus production can be predicted by the inequality $y \leq 26x - 6,600$, where y is Florida's annual production of citrus in thousands of tons and x is the number of thousands of acres of land.
 - a. Select the statements that could be reasons why the inequality modeling the relationship between acreage and citrus production uses the symbol $\leq$.
 - ☐ The acreage of land that actually bears fruit is less than or equal to the total acreage.
 - ☐ Extreme weather and/or disease could result in a less abundant citrus crop that year.
 - ☐ Citrus trees may start to produce more as they get older.
 - b. Discuss the statements that you chose with your partner.

The boundary line of the inequality, $y = 26x - 6,600$, is shown on the graph.



- c. Discuss with your partner which region to shade to represent the inequality $y \leq 26x - 6,600$.
- d. Shade the graph of the inequality $y \leq 26x - 6,600$.
- e. The ordered pair (400, 1,000) is in the solution set of the inequality. Explain what this solution represents in the real-world context of citrus production.
- f. The ordered pairs listed are all in the solution set of the mathematical inequality. Identify which ordered pair is not a viable solution in the real-world context.
 - ☐ (300, 0)
 - ☐ (300, 1,000)
 - ☐ (300, -1,000)Explain your selection.
- g. Which constraint on the values of x and y makes the solutions viable?
 - ☐ Both x and y must be greater than 0.
 - ☐ Both x and y must be non-negative.
 - ☐ Both x and y must be integers.
 - ☐ Both x and y must be rounded to two decimal places.



6.6.6 Wrap-Up: Reflection

1. Draw an **X** on each line to indicate how you feel about each learning target.

- a. I can graph the solution set of a real-world linear inequality, given in standard form, slope-intercept form, or point-slope form on a coordinate plane.

I need help. I can't get started.	I'm getting there, but I need more practice.	I understand this and I feel confident.
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- b. I can interpret solutions to problems as viable or non-viable options.

I need help. I can't get started.	I'm getting there, but I need more practice.	I understand this and I feel confident.
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Unit 6, Lesson 7: Writing Two-Variable Linear Inequalities



Benchmark

MA.912.AR.2.7 Write two-variable linear inequalities to represent relationships between quantities from a graph or a written description within a mathematical or real-world context.



Learning Target

- ☐ I can write a two-variable inequality given the graph of a solution set.